

Pharmacogenetics

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Objectives:

- Definitions
- Brief review of fundamental aspects of pharmacology
- Polymorphism in pharmacogenetics
- Effect of genetic variation on pharmacokinetics and pharmacodynamics
- Personalized medicine



- **Pharmacogenetics** is used to describe the influence of genes on the efficacy and side effects of drugs.
- It refers to genetic differences in metabolic pathways which can affect individual responses to drugs, both in terms of **therapeutic** effect as well as **adverse effects**.



- **Pharmacogenomics:** describes the interaction between drugs and the genome (i.e., multiple genes)

The key distinction between pharmacogenetics and pharmacogenomics



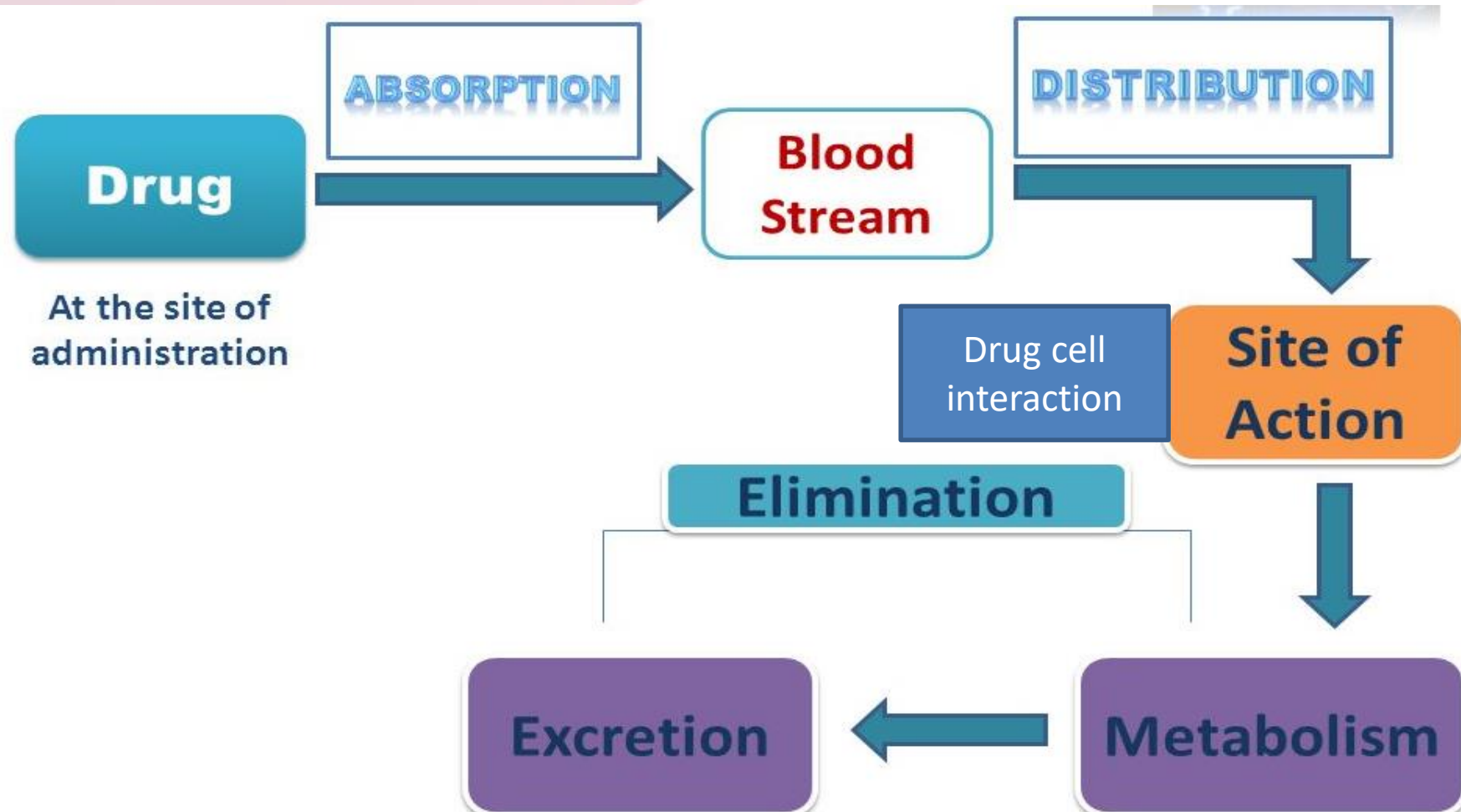
- **Pharmacogenetics** describes the study of variability in drug responses attributed to individual genes.
- **Pharmacogenomics**
the study of the entire genome related to drug response.

PRINCIPLES OF DRUG METABOLISM



- **Intake**
- **Absorption**
- **Distribution**
- **Drug cell interaction (receptor)**
- **Breaking down**
- **excretion**

PRINCIPLES OF DRUG METABOLISM



Biochemical modification of drugs



The actual **breakdown process**, which usually takes place in the **liver**, **varies** with different drugs.

- Some are oxidized completely to carbon dioxide, which is exhaled through the lungs.
- Others are excreted in modified forms via the kidneys into the urine
- Or excreted by the liver into the bile and hence the feces.

Biochemical modification of drugs



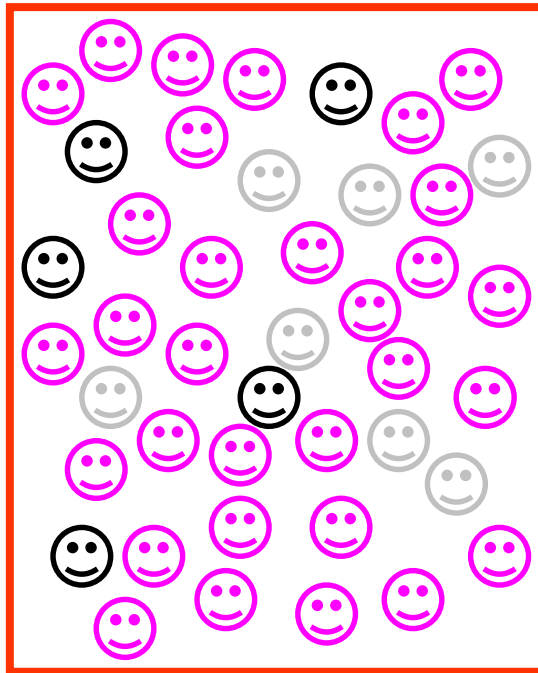
Many drugs undergo biochemical modifications that increase their solubility, resulting in their being more readily excreted:

- **conjugation**: union of carbohydrate (CHO)-glucuronic acid
e.g. Morphine , codeine
- **acetylation process**: introduction of acetyl group into the molecule
e.g. sulfonamide- Isoniazid

People react differently to drugs



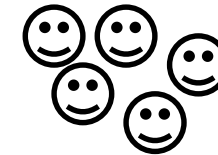
- ☹️ Toxic responders
- 😐 Non-responders
- 😊 Responders



Patient population with same disease phenotype

[https:// images.google.com](https://images.google.com)

Genotyping



Patients with drug toxicity



Patients with non-response to drug therapy



Patients with normal response to drug therapy



GENETIC VARIATION

- inherited variation at the DNA level results in functional variation in the gene products that play an essential role in determining the variability in responses, both therapeutic and adverse reaction to a drug.



GENETIC VARIATION

- Sequence Differences can Be in **Coding** Regions, **Regulatory** Regions (i.e., Promoters, Enhancers), Introns, at 5' or 3' Flanking Regions, etc



GENETIC VARIATION

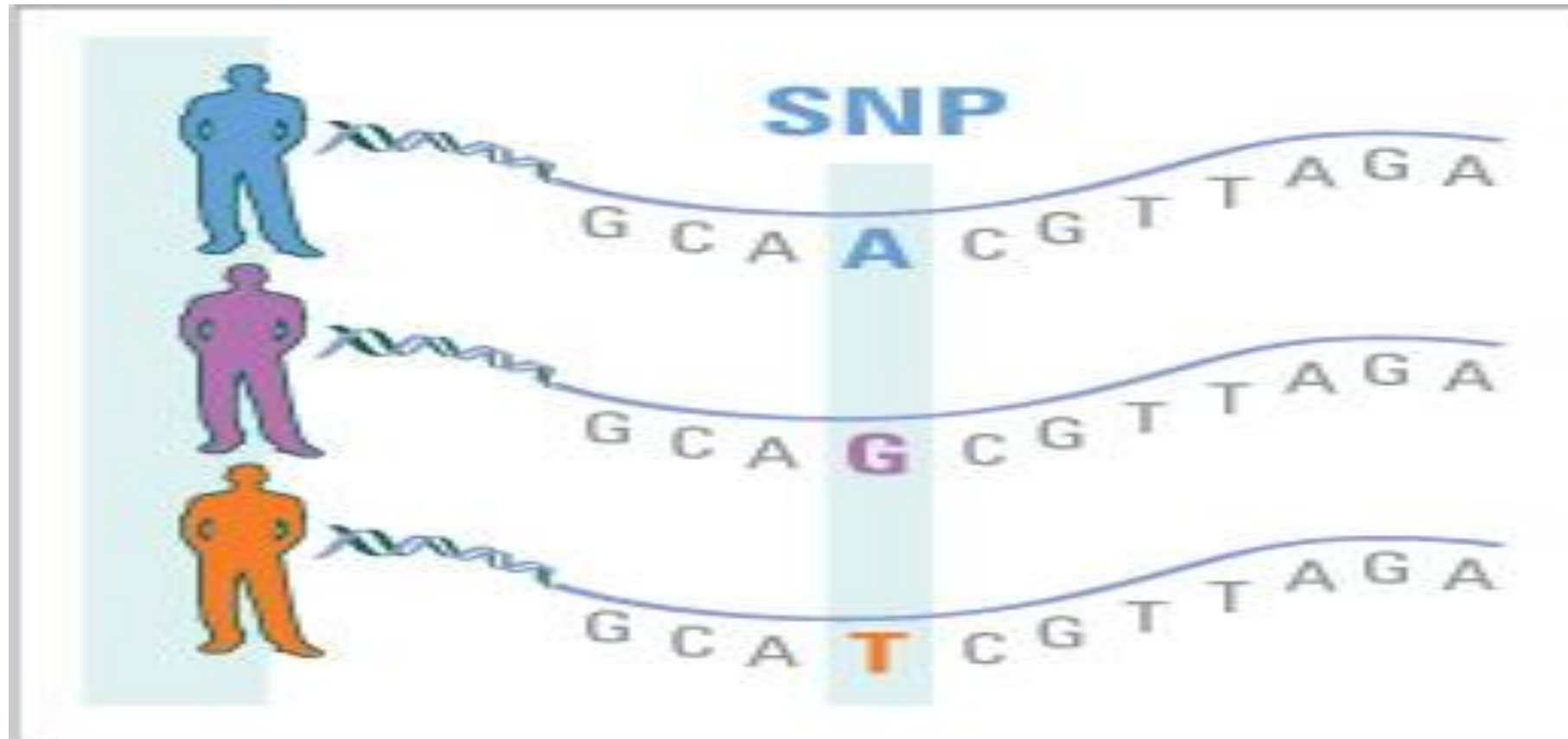
- If DNA sequence variation occurs in the coding portion or regulatory regions of genes, it is likely to result in variation in the gene product through alteration of function, activity, or level of expression.



GENETIC VARIATION

- **Polymorphism** , The occurrence in a population of two or more genetically determined forms of a gene each has a frequency of **MORE THAN 1%** of individuals of the given population

SNP: A single nucleotide DNA sequence variation





GENETIC VARIATION

- **Rare Variants** : the Variations Are Called rare variant If Alleles Are Present in $<1\%$ of Individuals in a Given Population,

Isoniazid (INH) inactivation: (N-Acetyltransferase activity)



It is a drug used in treatment of tuberculosis

It is rapidly absorbed from gut. ➡ Initial high blood level which slowly reduced as the drug is inactivated and excreted.

● Two groups of inactivators (acetylators):

- **Rapid inactivator**: The blood level of the drug falls rapidly after an oral dose.
- **Slow inactivator**: The blood level of the drug remains high for sometimes after an oral dose.

Isoniazid (INH) inactivation: (N-Acetyltransferase activity)



- Type of genes expressing NAT

NAT1: no difference between individual

NAT2: genetic variation responsible for polymorphism variation.

➤ **Slow inactivator (acetylators):**

Homozygous of autosomal recessive allele of liver enzyme N-Acetyltransferase (NAT2)

RAPID & SLOW ACETYLATORS



Individuals Who Are Rapid Acetylators (inactivator):

- Have **Higher Failure Rate With Isoniazid Therapy** for Tuberculosis
- Require **Larger Doses of Hydralazine** to Control Hypertension and **Dapsone** to Treat Leprosy and Other Infections
- **Liver damage**

RAPID & SLOW ACETYLATORS



Individuals who Are Slow Acetylators

Have high Risk of:

- Developing a Drug-Induced *Systemic Lupus Erythematosus*-Like Syndrome
- *Polyneuritis*>> it interact with **pyridoxine** leads to its deficiency. (numbness, tingling sensations of the hands and feet, anorexia and vertigo)
- *Hematological Adverse Reactions* After Isoniazid Treatment
- Increase risk of *developing number of cancers* (cancer bladder, colorectal, breast and lung)
- Side effects from *other drugs* e.g.
 - Hydralazin (antihypertensive)
 - Sulfasalazine (treatment of chron's disease)

Succinyl choline sensitivity (suxamethonium)



- It is used in anesthesia to produce skeletal and respiratory muscle relaxation with short period of apnea (to facilitate intubation).
- Metabolism: destroyed by pseudocholine esterase (CHE).
- Mutations in the *pCHE* gene
- About 1/2000 has a period of apnea that can last 1 hour or more after the use of Succinylcholine.

Succinyl choline sensitivity (suxamethonium) AR



- When a succinylcholine-induced apnea occurs the anesthetist has to maintain respiration until the effects of the drug have worn off

Certain variants in pCHE gene

- In patient highly sensitive to it : CHE slowly destroy Succinyl choline slower than normal.
- Complete absence of CHE prolonged apnea.

Glucose-6- Phosphate dehydrogenase deficiency (G-6-PD) deficiency



- G-6-PD involved in pathway of red cell glycolysis (hexose monophosphate shunt) to maintain Glutathion in the reduced state to maintain the stability of the RBCs in the presence of certain oxidizing drugs.

Glucose-6- Phosphate dehydrogenase deficiency (G-6-PD) deficiency



- **Mutation:** Point mutation in G6PD gene
 - **Defects:** decrease production of enzyme
or normal production with decrease catalytic activity.
- Membrane Fragility → Hemolysis → Hemolytic Anemia

Glucose-6- Phosphate dehydrogenase deficiency (G-6-PD) deficiency



- G6PD deficiency is inherited as an **X-linked recessive trait**
- rare in Caucasians but **affecting approximately 10% of Afro-Caribbean males and relatively common in the Mediterranean.**
- It is thought to be relatively common in these populations as a result of conferring increased resistance to the malarial parasite.

Glucose-6- Phosphate dehydrogenase deficiency (G-6-PD) deficiency



Clinically:

clinical picture appear after exposure to certain drugs :

- ☐ very dark, often black urine.
- ☐ Jaundice
- ☐ Red cell count and hemoglobin concentration gradually fell as a consequence of hemolysis of the red blood cells.

It can be fatal

Glucose-6- Phosphate dehydrogenase deficiency (G-6-PD) deficiency



Drugs cause hemolysis:

- Antimalarial drugs e.g. (primaquine, nitrofurantoin, phenacetin), Asprine, and certain sulfonamides.
- sulfa drugs,
- foods (fava beans)
- infection

Coumarin Metabolism



- Anticoagulant drug: prevent blood clotting.

such as : warfarin

Uses: treatment of deep venous thrombosis(DVT)

Gene:**CYP2C9** gene (encode for the cytochrome P450 enzyme)

Two variants(CYP2C9*2 and CYP2C9*3 genes) result in decreased metabolism

Patients increased risk for bleeding

A lower warfarin dose used to maintain their target international normalized ratio (INR) range.

Thiopurine methyltransferase



- Thiopurines are a group of toxic substances used for **suppression** of **immune response** in patients with:
 - ✓ autoimmune disorders e.g. systemic lupus erythematosus
 - ✓ treatment of leukemia.
 - ✓ to prevent rejection of organ transplants
- e.g. 6- mercaptopurine, 6- thioguanine and **azathioprine**.
- They have serious side effects:
 - leukopenia
 - severe liver damage.

Thiopurine methyltransferase



- Azathioprine is reported to cause toxicity in 10-15% of patients.
- two-thirds of patients who experience toxicity have one or more variant alleles in TPMT gene
- This gene encode for Thiopurine methyltransferase:
enzyme responsible for methylation of Thiopurines .

Debrisoquine metabolism



- Debrisoquine :drug that was used for **treatment of hypertension**, in the past
- Gene involved in metabolism: **CYP2D6** one of P450 family on chromosome 22.

Defect:

- certain variants lead to: reduced hydroxylation activity.
- Poor metabolizers :5-10% of persons of European origin (British) homozygotes for an autosomal recessive gene with reduced hydroxylation activity.

Complain: **overactivity of the drug leads to hypotension.**

Debrisoquine metabolism



- CYP2D6 variation is important because this enzyme is **involved in the metabolism** of more than **20% of prescribed drugs**, e.g.
- B-blockers : propranolol and metoprolol, amitryptiline, imipramine, and tamoxifen.

Dihydropyrimidine Dehydrogenase (DPYD)



- Rate-limiting enzyme in the **catabolism** of the chemotherapeutic drug **5-fluorouracil (5FU)**.
- **Deficiency of DPYD** : severe 5FU-associated toxicity.
- **Measurement of DPYD** activity in **peripheral blood** mononuclear cells or **genetic testing** for the most common **DPYD gene** variations may be warranted in cancer patients before the administration of **5FU**.

Malignant hyperthermia



- A rare complication of anesthesia
- C/P
 - **Hyperthermia** (increase temperature), up to 42.5 degree.
 - **Muscle rigidity** during anesthesia

Occur when **Halothan** used in anesthesia especially when **Succinylcholine** used as muscle relaxant.

Treatment by **vigorous cooling**

Prediction of individual susceptibility :

In vitro muscle biopsy and testing the response to Halothan and caffeine.(muscle contraction)

Malignant hyperthermia



- ❖ **MH susceptibility** is inherited as an **autosomal dominant** trait affecting approximately 1 in 10,000 people.
- ❖ MH is genetically heterogeneous, but the most common cause is a mutation in the **ryanodine receptor (RYR1) gene**.
- ❖ Variants in other genes may influence susceptibility

Personalized Medicine



- Using genetic or genomic information to select the most appropriate choice of pharmacological therapy at the correct dosage for every person.

Maturity-Onset Diabetes of the Young



- Monogenic form of diabetes characterized by **young age of onset** (often before the age of 25 years).
- AD
- **Beta cell dysfunction.**
- Many patients are **misdiagnosed** with type 1 diabetes and treated with insulin.

Maturity-Onset Diabetes of the Young



- HNF1A gene mutation causing MODY
- There are fourfold increased response to sulfonylureas in patients with HNF1A mutations compared with a control group with type 2 diabetes

Maturity-Onset Diabetes of the Young



- For many patients a genetic diagnosis of HNF1A MODY means that they can transfer from insulin injections to sulfonylurea tablets.

Neonatal Diabetes



- The most frequent cause of permanent neonatal diabetes is an activating mutation in the **KCNJ11 or ABCC8 genes**, which encode the Kir6.2 and SUR1 subunits of the ATP sensitive potassium (K-ATP) channel in the pancreatic beta cell.
- The effect of such mutations is to prevent K-ATP channel closure by reducing the response to ATP.
- Because channel closure is the trigger for insulin secretion, these mutations result in diabetes, and a requirement for lifelong insulin treatment.

Neonatal Diabetes



- Defining the genetic etiology for this rare subtype of diabetes has led to improved treatment, because most patients can be treated successfully with sulfonylurea tablets instead of insulin.
- These drugs bind to the sulfonylurea receptor subunits of the K-ATP channel to cause closure independently of ATP, thereby triggering insulin secretion

Neonatal Diabetes



- High-dose sulfonylurea therapy results in improved glycemic control which will reduce the risk of diabetic complications in later life.

Neonatal Diabetes



- International guidelines now recommend genetic testing for anyone diagnosed with diabetes in the first 6 months of life in order to identify those patients who will benefit from sulfonylurea treatment



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THANK YOU